

Sumedh Kothari

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EDUCATION

Georgia Institute of Technology

B.S. in Chemical and Biomolecular Engineering

Minor in Computer Science

Expected May 2028

GPA: 3.82

- **Coursework:** Computer Organization and Programming; Computing for Engineers; Advanced Data Structures, Embedded Systems, and Networking; Ordinary Differential Equations
- **Honors:** Symposium of Rising Scholars Presenter, Eagle Scout
- **Organizations:** American Institute of Chemical Engineers, Technical Association of the Pulp and Paper Industry

EXPERIENCE

Georgia Tech Renewable Bioproducts Institute | Tong Laboratory

Undergraduate Research Assistant

Atlanta, GA

August 2025 – Present

- Used Python, NumPy/Pandas libraries to apply Bayesian statistical modeling to optimize lignin valorization reaction
- Utilized NMR, SEM, and HPLC modeling data to characterize value-added chemicals from lignin breakdown
- Completed comprehensive literature reviews on aromatic compound synthesis, 3D printing custom GO structures, and optimal catalyst design while collaborating with research team to write manuscript for publication

ChemE Cube | Georgia Tech

Team Lead – Mechanical

Atlanta, GA

August 2025 – Present

- Leading the mechanical development of a 1 ft³ direct air capture reactor by managing a 5-member subteam and overseeing end-to-end CAD, prototyping, and systems integration, delivering a fully fabricated and test-ready prototype
- Designing airflow distribution and sorbent containment subsystems using parametric CAD, 3D printing, and CNC machining, achieving 75% reduction of ambient CO₂ in bench-scale validation and preparing the system for AIChE competition (Nov 2026)

Dressen Laboratory

Researcher

Los Altos, CA

August 2024 - June 2025

- Designed, tested fluorinated Mg₂dobpdc metal organic frameworks for vehicle exhaust CO₂ capture, improving capture potential by 20% while improving stability by 115%
- Characterized adsorption efficiency and regeneration performance using TGA, NMR, and cyclic adsorption experiments, with results presented at SYNOPSYS 2025

PROJECTS

IEEE Robotech 2026 Hackathon - Autonomous Track | Georgia Tech

Mechanical/Software Engineer

Atlanta, GA

January 2026

- Designed and CAD-modeled a fully autonomous debris-clearing robot in SolidWorks by engineering the chassis, actuation mechanisms, and beacon-deployment system, enabling end-to-end autonomous navigation and operation
- Developed and deployed a YOLO-based computer vision model in PyTorch by training and integrating it into the onboard perception pipeline, achieving 95% debris detection accuracy and enabling real-time hazard identification in landing zones
- Integrated perception, navigation, and mechanical actuation systems by linking CV outputs to motion planning and debris-removal hardware, allowing the robot to autonomously clear obstacles and deploy transponder beacons to certify landing

EcoFlow - Sustainability Calculator | Python, FastAPI, Next.js, Redis, PostgreSQL

Developer

Atlanta, GA

January 2026

- Built and deployed a FastAPI (Python) + Next.js (TypeScript) full-stack web app that processes any Amazon ASIN across 5 lifecycle stages, automating cradle-to-grave LCA calculations and reducing analysis time from hours to seconds via AI-driven product parsing and structured emissions modeling
- Engineered asynchronous REST APIs with Redis caching (24-hr TTL) and PostgreSQL persistence, containerizing services with Docker to achieve <1s repeat-query latency while improving scalability and cost efficiency

AiApply - AI Job Applier

Developer

Atlanta, GA

October 2025

- Built full-stack AI internship application platform automating resume optimization, cover letter generation, and programmatic job submissions via Handshake, streamlining end-to-end job search workflows
- Developed Selenium automation pipeline to scrape recruiter profiles from filtered company searches and deploy personalized DM outreach at scale, significantly increasing outreach throughput and response rates
- Engineered web-scraping and targeting algorithms to identify companies by location/industry filters and trigger automated, resume-aware email outreach with dynamically generated candidate skill summaries

SKILLS

- **Technical:** Python, FastAPI, C++, Next.js, Redis, PostgreSQL, Data analysis (Excel, MATLAB), CAD (SolidWorks)
- **Communication:** Technical Writing, Multilingual (English, Hindi, Spanish)